2023 7th International Conference On Computing, Communication, Control And Automation (ICCUBEA)
Pimpri Chinchwad College of Engineering (PCCOE), Pune, India. Aug 18-19, 2023

Multi-Facial Automated Attendance System using Haar Cascade, LBPH, and OpenCV-Based Face Detection and Recognition

Snehal Rathi
Vishwakarma Institute of Information Technology
Pune, India
snehal.rathi@viit.ac.in

Snehal Kapadnis
Vishwakarma Institute of Technology
Pune, India
snehal.kapadnis20@vit.edu

Gayatri Sonar
Software Engineer, Aress Software,
Nashik, India
gayatri9213@gmail.com

Gaurav Sonar
Vishwakarma Institute of Information Technology
Pune, India
gaurav.22020272@viit.ac.in

Sushma R. Vispute
Pimpri Chinchwad college of Engineering
Pune, India
sushma.vispute@pccoepune.org

2023 7th International Conference On Computing, Communication, Control And Automation (ICCUBEA) | 979-8-3503-0426-8/23/$31.00 ©2023 IEEE | DOI: 10.1109/ICCUBEA58933.2023.10392248

***Abstract*- In the 21st century, everything around us is dependent upon technology to make our lives much easier. Daily tasks are increasingly being computerized. Nowadays more people prefer to do their work electronically. To the extent of our knowledge, students' attendance at some universities is manual. The attendance management system computerizes the traditional attendance process. Utilizing OpenCV (Open-source Computer Vision) and some face detection and recognition algorithms, we propose a system to measure student attendance in classes. By using face detection algorithms and face recognition algorithms, we will keep track of how many people are present in the class and attendance of each one. This project intends to provide a system that automates and streamlines the process of collecting and monitoring student attendance using facial recognition technology. To find, identify, and validate collected faces, pre-processing methods like the Haar Cascade approach will be used. To make the attendance process quicker and more accurate, we aim to offer a method.**

Keyword: attendance management system, computer vision, face detection, face recognition, Haar Cascade approach,

I. INTRODUCTION

The procedure for recording student attendance has evolved and changed during the last ten years. The need to automate, simplify, accelerates, and save time and effort is what propels this progress. The teacher calls out for each student and individual to mark their attendance while using the manual attendance method, even though attendance systems are presents everywhere. Even if it is a time-consuming process, there are also chances of having manual errors. Also, one of the main problems teachers could have in class is low attendance. In this project, we try to cut down on idling time, get rid of mistakes, and automate the procedure. Our method uses facial recognition technology to track attendance using a high-resolution camera that can detect and identify the faces. It then compares the identified faces with students face images recorded in a database. An individual's attendance is noted in the attendance database once their face matches a saved photograph. Consequently, the focus of our project will be on online student attendance. In other words, the objective of this paper is to create a system which enables professors to take student attendance in a accurate automated manner.

II. PROBLEM STATEMENT

Lecturers that use the current attendance system find issues such as manually recording student attendance, messing up attendance papers, dishonesty, lost time, and high loss rates. It is a difficult process that takes time and involves a lot of paperwork. So, to address these issues and prevent mistakes, we recommend computerized procedure by offering a software program that automatically tracks and monitors student attendance without the involvement of teachers.

III. LITERATURE REVIEW

The proposed system addresses the issue of proxy attendance and students being marked as present even when they are not present physically. The system makes use of live video stream and from the video frames are extracted using OpenCV library. It uses dlib library to detect the faces and recognize them. After these, by comparing the actual faces with the database which contains students' faces, the attendance of respective student is marked successfully [1].

The Proposed system is implemented in four phases: Image Capturing, Image Segmentation from group image and based on that image Face Detection will be done [8], Face comparison will be done against database, Lastly Attendance will get update accordingly. The objectives are to save time and make system simple and easy to use [2].

The Attendance Management Using Facial Recognition enrolls the faces of student subject wise into the database against the subject ID and Name[12]. The system then allots attendance to the recognized faces in the database [3].

Several techniques are discussed for detecting the face, such as the Ada-Boost algorithm for detection of the faces; the Viola Jones algorithm for face detection, SNOW Classifier method and the SMQT features, and Local Binary Pattern (LBP) are used [4].

Students or staff must wait for the completion of the previous members to prevent proxy issues and maintain attendance effectively [9]. The proposed system detects and recognizes iris patterns by having the student stand in front of a camera. In order to detect the iris, some algorithms are used, such as Gray Scale Conversion, Rectangular Filter, and Skin Pixel Detection [5].

Authorized licensed use limited to: The Technology Library. Downloaded on July 13,2025 at 17:53:19 UTC from IEEE Xplore. Restrictions apply.

Face recognition has been improved through numerous algorithms and techniques, but Deep Learning is being applied here [11][13]. This tool converts the frames of the video into images in order to identify the face of the student so that their attendance can be automatically reflected in the attendance database [6].

Face detection rate is not high enough to allow us to estimate attendance precisely based on individual face recognition results[10]. In this paper author propose the method to estimate attendance precisely using continuous observation which improves the accuracy of attendance[14]. The system uses raspberry pi with OpenCV library. Thus, the attendance approximation performance will be improved due to uninterrupted observation [7].

A. *Existing System:*

Lecturers still utilize the manual attendance system, which involves calling out students' names or circulating an attendance sheet among them so that each student can sign it next to their name. Both methods take a lot of time and have large error margins. There are many issues with collecting attendance in this manner traditionally, as students occasionally fill in for friends or teachers who are absent.

B. *Proposed System:*

By taking a picture of the entire class and uploading it to a system that detects and recognizes the students' faces, facial recognition technology is used to track attendance. The system then matches the identified face with student face photographs that are kept in the database. Based on the result of comparison, attendance of that student for respective subject is marked.

IV. OBJECTIVE

Our main objective is to support the teachers in their efforts to track and control student attendance and absenteeism by streamlining and organizing the process. In addition, using this system,

- Teachers can keep track of defaulters and take suitable action at the correct time.
- Students can keep track of their attendance.
- By offering an automatic and trustworthy attendance system which makes use of facial recognition technology, manual process mistakes can be reduced.
- It boosts privacy and security so that students can't introduce themselves or their friends when they aren't there.
- Being adaptable and having the flexibility to change attendance recordsfor lectures.

Features

- The device automatically records attendance and retains the faces that are detected.
- Allow authorized access.
- Ease of use.
- Several faces detected at a time.

V. METHODOLOGY

A. *Face detection*

In many applications, face detection is a technique for recognizing human faces in digital images. At the process of face detection, faces will be found in captured photographs so that they can be compared. The Viola-Jones method is used to detect faces.

Haar Cascade Algorithm: This is commonly used algorithm for object detection. This algorithm's key characteristic is that detection happens quickly while training takes a while. Generating the integral image first can greatly improve the performance of this technique.

- Haar features.
- Integral Image
- Cascading

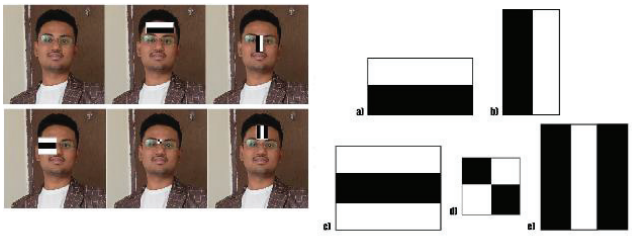


Fig. 1. Haar Feature on Training Image

1) *Face Recognition*

The face recognition algorithm oversees determining the features that best define the image once the facial images have been retrieved, cropped, scaled, and typically converted to grayscale. Face recognition algorithms come in a variety of forms, such as Eigen Faces, LBPH (Local Binary Patterns Histograms), and Fisher Faces. For the suggested system, LBPH is the most appropriate of these algorithms.

B. *LBPH Algorithm:*

A straightforward yet very effective texture operator, the Local Binary Pattern (LBP), labels pixels a picture by thresholding the area around each pixel. Steps involved in

- Radius, Neighbors, Grid Parameter Xand Y.
- Practicing the LBP operation.
- Extracting the histograms.
- Training the algorithm
- Using face recognition

C. *OpenCV*

One of the objectives of OpenCV is to provide a simple-to-use computer vision infrastructure that allows the development of sophisticated vision applications. One of OpenCV's goals is to provide a computer vision infrastructure that is easy to use so that people can build sophisticated vision applications quickly. Face detection, feature extraction, and face recognition are the three stages of a facial recognition system, as shown in Figure.

Authorized licensed use limited to: The Technology Library. Downloaded on July 13,2025 at 17:53:19 UTC from IEEE Xplore. Restrictions apply.

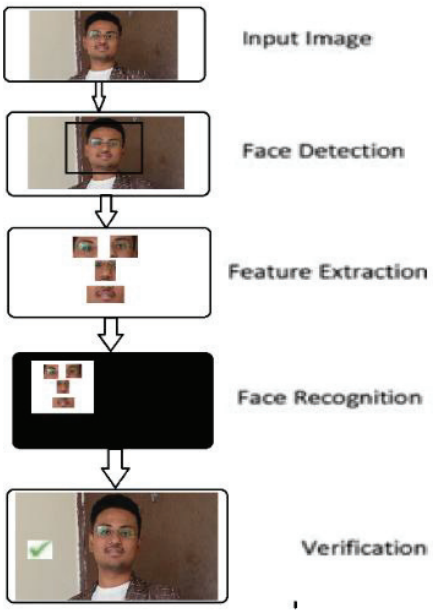


Fig. 2. Structure of face recognition system

VI. IMPLEMENTED SYSTEM

We propose a system that finds and recognizes faces using the Haar Cascade object detector method. This may be accomplished using Open CV's in-built Haar cascades. Trained faces are first entered into a database, and the trained photos are then compared (trained images are initially available in the database indicates the linked folks to the specific college or organization). That will result in attendance to the people who have been identified after comparison.

A. Faculty Module:

In this section, there is functionality like add faculty, in which we can register new faculty members in our domain by taking some required details and store these details in our database. There will be options also like department, class, or division wise we can show all the faculty members.

B. Student Module:

The student module works like the faculty module. It will also have functionality of register student, modify details of student, displaying all the students present in domain. But while registering students we are taking input images of students which will be work for our attendance module.

C. Subject Module:

In this Subject tab, there will be subject registration and subject allocation. Here we can add subject related details and then allocate subjects to the faculty member. From this faculty will understand the subject or class for which they need to mark the attendance. According to that faculty will mark the attendance of different subject lectures.

D. Attendance Module:

The attendance module is the most important part or main objective of this project. So, in this, we are marking the attendance of students which are present at the time of lecture or any class. Here we must upload the images of students. Once the model is trained using algorithms system detects and recognizes the face. After comparing the image with the database's trained images, the algorithm records attendance for recognized faces. Also, users can view the attendance of present students from View Attendance tab. So, like this, it will work very efficiently and in user friendly manner.

VII. ARCHITECTURE DIAGRAM

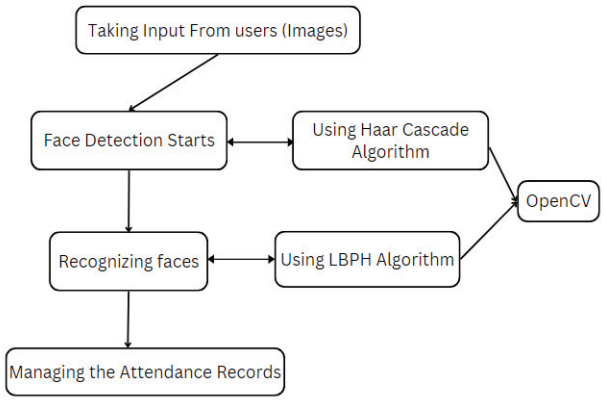


Fig. 3. Architecture Diagram

VIII. RESULTS AND DISCUSSION

A. Image Upload:

Firstly, we need to register students into the database and upload image while registering student. The image will store in folder with name of student ID.

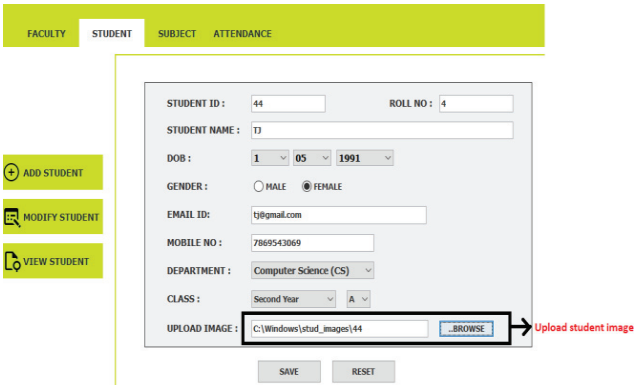


Fig. 4. Primary Faces Registration

B. Face Detection:

The image is used as an input for face detection, which looks for any faces. Once the face is located, the image is cleaned up for easier recognition. Facial recognition can be achieved by implementing the Haar Cascade algorithm. Datasets are trained using Haar Cascade to obtain unique features of their face (eyes, noses, ears, lips). It works by giving us most important parts of the face for detection through Haar cascade algorithms.

C. Face Matching

The LBPH algorithm (Local Binary Pattern Histogram) is used to compare the captured images with the stored images in the database to recognize the student. Multiple student faces are detected by the system and according to face recognition attendance will be marked and stored into database. A single instance will be able to handle at least 15 faces.

Authorized licensed use limited to: The Technology Library. Downloaded on July 13,2025 at 17:53:19 UTC from IEEE Xplore. Restrictions apply.

D. *Fetch Attendance*

Students or Faculty can fetch attendance by proving appropriate data like Department, Date and for which subject they want to fetch attendance of present students.

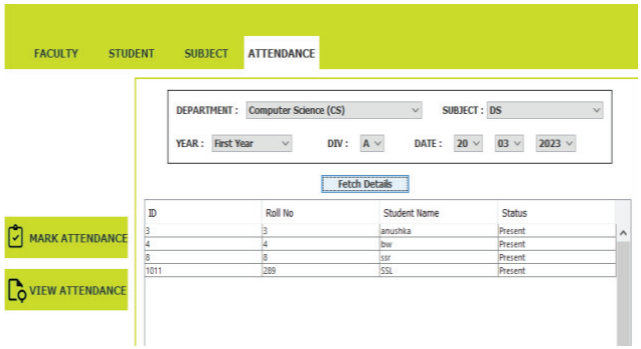


Fig. 5. Fetching the attendance fromdatabase.

Advantages

- This system can be applied to secured areas and organizations for security purposes.
- The software automatically records attendance while storing the observed faces.
- Technology offers users convenience and security.
- That helps them save time and effort.

Disadvantages

- The system doesn't recognize faces well in low light, which could lead to erroneous results.
- It is only capable of detecting close up faces.

IX. CONCLUSION

Hence, a system that is trustworthy, secure, quick, and efficient has been devised to replace a manual and unreliable attendance system. Face detection using Haar-Cascades performed incredibly well. A cost-effective face recognition platform can be created by combining LBPH and Haar-Cascades. For better outcomes in the management of attendance, this approach can be put into place. The technology will save time, lessen the amount of work that needs to be done by the administration, replace paper documents with digital ones, and cut down on the number of people needed for the job.

REFERENCES

[1] Hl, D.G., Vishal, K., Dubey, N.K. and Pooja, M.R., 2020. Face recognition-based attendance system. International Journal of Engineering Research & Technology (IJERT), 9(6).

[2] Gomes, C., Chanchal, S., Desai, T. and Jadhav, D., 2020. Class attendance management system using facial recognition. In ITM Web of Conferences (Vol. 32, p. 02001). EDP Sciences.

[3] Bharadwaj R., Rao T., R. 2020. Attendance Management Using Facial Recognition. International Journal of Innovative Technology and Exploring Engineering (Vol-8 Issue-6,).

[4] Gupta, V. and Sharma, D., 2014. A study of various face detection methods. International Journal of Advanced Research in Computer and Communication Engineering, 3(5), pp.6694-6697.

[5] Singh D., Kusuma M., Mounika K. 2019. IOT Based Automated Attendance with Face Recognition System. International Journal of Innovative Technology and Exploring Engineering (Vol-8, Issue-6S4).

[6] Nandhini, R., Duraimurugan, N. and Chokkalingam, S.P., 2019. Face recognition-based attendance system. International Journal of Engineering and Advanced Technology (IJEAT), 8(3S)

[7] Katara, A., Kolhe, M.S.V., Zilpe, M.A.P., Bhele, M.N.D. and Bele, M.C.J., 2017. Attendance system using face recognition and class monitoring system. International Journal on Recent and Innovation Trends in Computing and

[8] Communication, 5(2), pp.273-276

[9] Kalyankar, P.A., Mulani, A.O., Thigale, S.P., Chavhan, P.G. and Jadhav, M.M., 2022. Scalable face image retrieval using AESC technique. Journal Of Algebraic Statistics, 13(3), pp.173-176

[10] Kaur, H., & Singh, A. (2020). An intelligent attendance management system using face recognition. In 2020 7th International Conference on Signal Processing and Integrated Networks (SPIN) (pp. 156-161). IEEE.

[11] Harshad Patil, Heena Khan, Saquib Ansari, Saimohan Sahu., 2021. Automatic endance System Using Face Recognition. In International Journal of Advanced Research in Science, Communication and Technology (IJARSCT) (Volume 4, Issue 1) SSN (Online) 2581-9429.

[12] S. R. Rathi and Y. D. Deshpande, "Embedding Affect Awareness into Online Learning Environment using Deep Neural Network," 2019 5th International Conference On Computing, Communication, Control And Automation (ICCUBEA), Pune, India, 2019, pp. 1-6, doi: 10.1109/ICCUBEA47591.2019.9128811.

[13] Ms. Dyanila Neeta Ferrao, Ms. Kusuma D R, Mr. Manoj Kammar, Mr. Pradeep N M, Mr. Vishal Hegde.,2022. Student Attendance System Using Face Recognition. IJRTI. (Volume 7, Issue 6), ISSN: 2456-3315

[14] S. R. Rathi and Y. D. Deshpande, "Embedding Affect Awareness into Online Learning Environment using Deep Neural Network," 2019 5th International Conference On Computing, Communication, Control And Automation (ICCUBEA), Pune, India, 2019, pp.1-6,doi: 10.1109/ICCUBEA47591.2019.9128811.

[15] S. Rathi, Y. Deshpande, S. Nagaral, A. Narkhede, R. Sajwani and V. Takalikar, "Analysis of User's Learning Styles and Academic Emotions through Web Usage Mining," 2021 International Conference on Emerging Smart Computing and Informatics (ESCI), Pune, India, 2021, pp. 159-164, doi: 10.1109/ESCI50559.2021.9397037.

Authorized licensed use limited to: The Technology Library. Downloaded on July 13,2025 at 17:53:19 UTC from IEEE Xplore. Restrictions apply.